



CCS2305:WEB PROGRAMMING

LECTURE #2

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OUTLINE



HTML Elements



Document Structure



Basic Tags



Attributes



Hyperlinks



Text formatting



Organizing Content

HTML ELEMENTS AND TAGS

- **Elements** are identified by tags in the text source. A **tag** consists of the element name within angle brackets (**< >**).
- Most of **HTML tags** are closed tags, which means they have a **start tag** and an **end tag**.
- For example, they look like the following:
- **<element-name>** content goes here... **</element-name>**.



Start Tag or Opening Tag



End Tag or Closing Tag

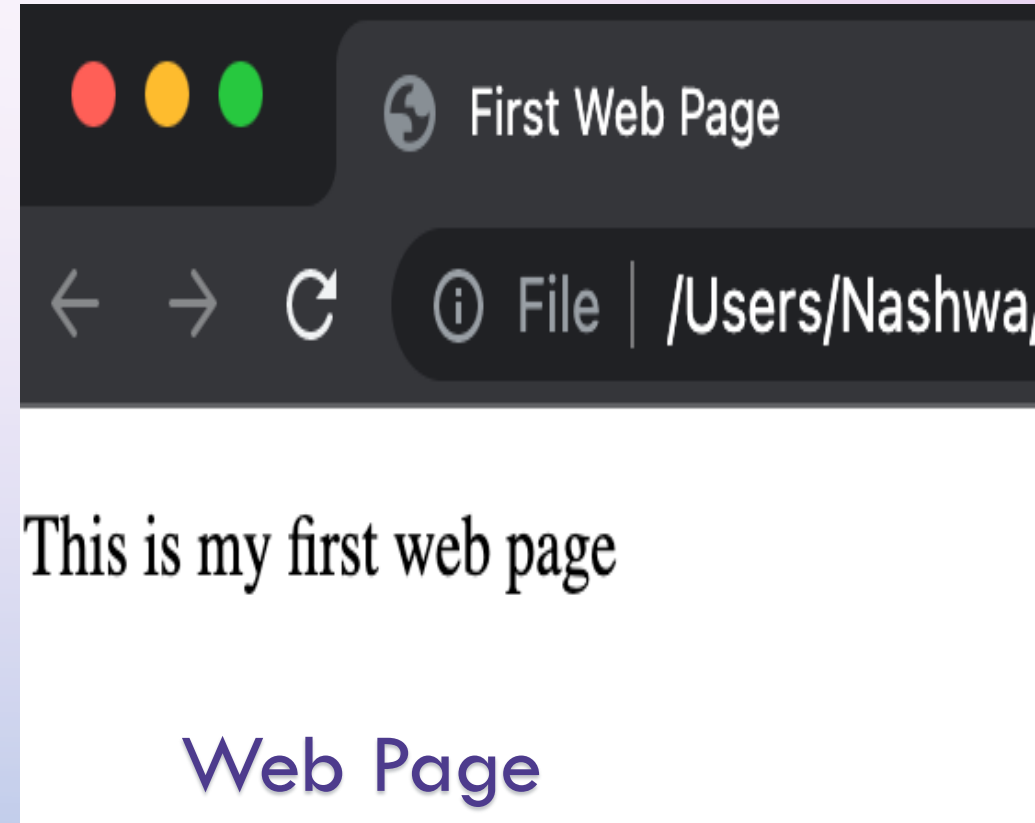
- The tags that do not have an end tag are called open tags. This means that they have no content to affect.

DOCUMENT STRUCTURE

firstwebpage.html

```
1  <!DOCTYPE html>
2  <html>
3    <head>
4      <title> First Web Page </title>
5    </head>
6    <body>
7      <p> This is my first web page </p>
8    </body>
9  </html>
```

Source Code



HTML DOCUMENTS

- The `<!DOCTYPE>` declaration must be the very first thing in your HTML document.
- The `<!DOCTYPE>` declaration is **not an HTML tag**; it is an instruction to the web browser about what **version of HTML** the page is written in.
- The `<!DOCTYPE>` specifies the rules for the markup language, so that the browsers render the content correctly.
- The `<!DOCTYPE>` declaration for HTML5 is `<!DOCTYPE HTML>`.

BASIC TAGS

- **Entire document** is enclosed between `<html>` and `</html>`.
All html documents begin with this tag, in which all other tags are contained.
- **Head section** is enclosed between `<head>` and `</head>`.
Contains information that the browser uses to control the look of the page.
Can also contain a title for the browser window, enclosed between `<title>` and `</title>`.
- **Body section** is enclosed between `<body>` and `</body>`.
Contains the text that will appear in the page.

HEAD ELEMENT

- The **head** element contains elements that pertain to the document that are not rendered as part of the content, such as its title, style sheets, scripts, and metadata.
- **Meta** elements provide document **metadata**, information about the document. In this case, it specifies the **character encoding** (a standardized collection of letters, numbers, and symbols) used in the document as Unicode version UTF-8
- Other types of metadata provided by the **meta** element are the author, keywords, publishing status, and a description that can be used by search engines.

`<meta charset="UTF-8">` OR

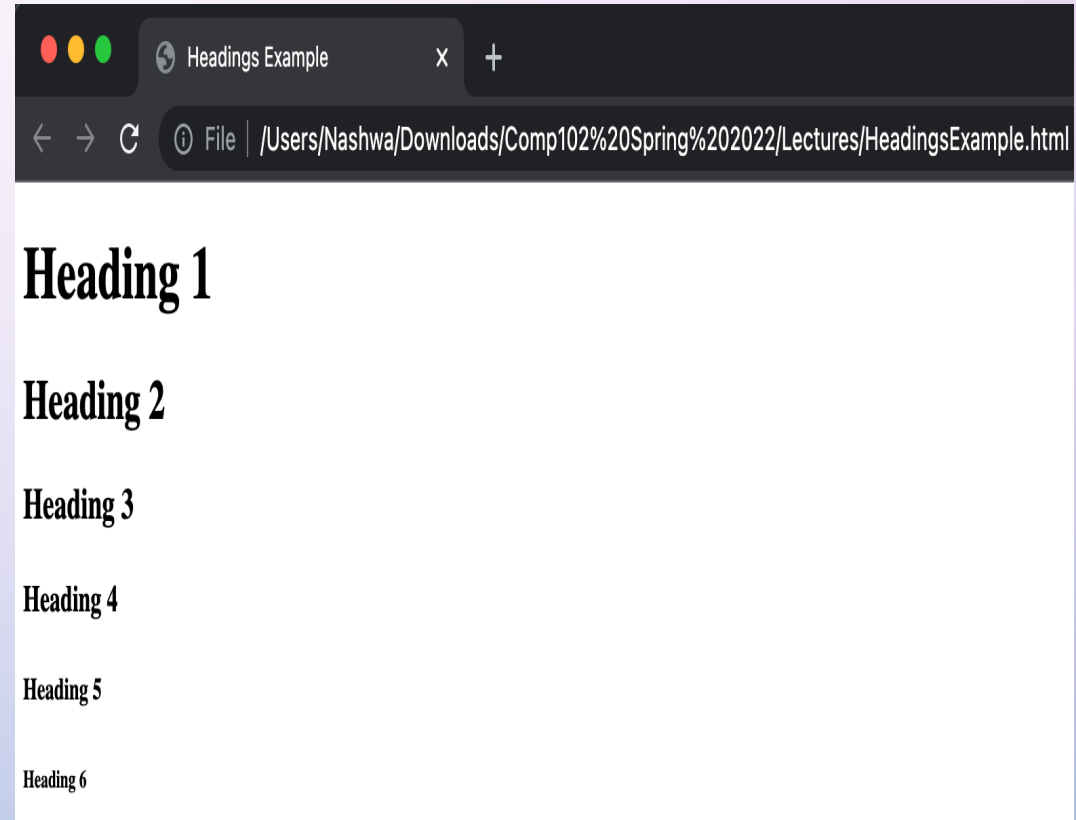
`<meta name="keyword" content="HTML, JavaScript">`

HEADINGS

- **Headings** are used to display **titles and subtitles** and structure the content of web pages.
- Browsers automatically add some white space (a margin) before and after a heading.
- HTML headings are defined with the **<h1>** to **<h6>** tags.
- **<h1>** headings should be used for main headings, followed by **<h2>** headings, then the less important **<h3>**, and so on.

HEADINGS EXAMPLE

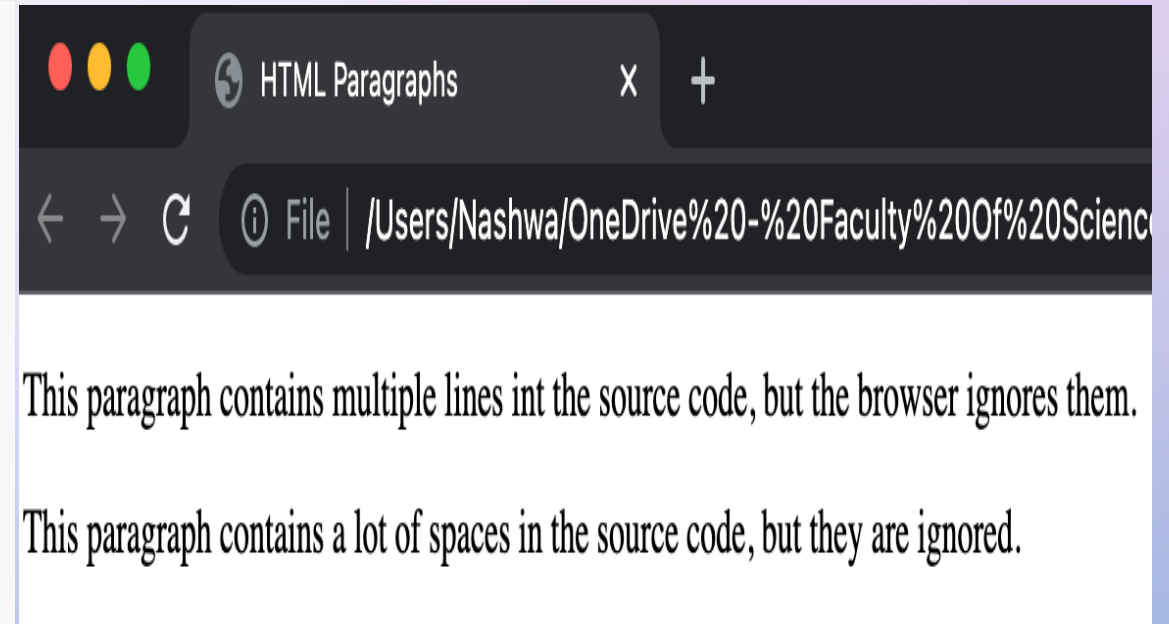
```
1  <!DOCTYPE html>
2  <html>
3  <head>
4  <title>
5      Headings Example
6  </title>
7  </head>
8  <body>
9      <h1> Heading 1 </h1>
10     <h2> Heading 2 </h2>
11     <h3> Heading 3 </h3>
12     <h4> Heading 4 </h4>
13     <h5> Heading 5 </h5>
14     <h6> Heading 6 </h6>
15 </body>
16 </html>
```



PARAGRAPHS

- Paragraphs are a block of text and usually start with a new line.
- HTML paragraphs are defined with the `<p>` tag.

```
<body>
  <p>
    This paragraph
    contains multiple lines
    int the source code,
    but the browser ignores them.
  </p>
  <p>
    This paragraph
    contains          a lot of spaces
    in the source    code,
    but they are     ignored.
  </p>
</body>
```



HTML TAGS

- The **
** tag is an empty element, which defines a **line break**.
- The **<hr>** tag is also an empty element and is used to separate content in an HTML page by displaying a **horizontal rule**.
- The **hr** element adds a logical divider between sections of a page or paragraphs without introducing a new heading level.
- The **<pre>** is a closed tag, which defines a **pre-formatted text**. This means preserving both spaces and line breaks.
- For content in which whitespace is semantically significant, use the preformatted text (**pre**) element. It is a unique element in that the content is displayed exactly as it is typed

COMMENT ELEMENT

- The comment element `<!-- comment here -->` is used to document pages and is not an HTML element. So, it is ignored by the browser.

CHARACTER ESCAPING

- Characters that might be misinterpreted as code need to be **escaped** in the source document.
- Escaping means that instead of typing in the character itself, you represent it by its numeric or named **character entity reference**.
- All character references begin with an **&** (ampersand) and end with a **;** (semicolon)
- Some characters have no graphic display and are difficult to see in the markup. These include the non-breaking space (** **), which is used to ensure that a line doesn't break between two words.
- Another use for non-breaking spaces is to separate digits in a long number,

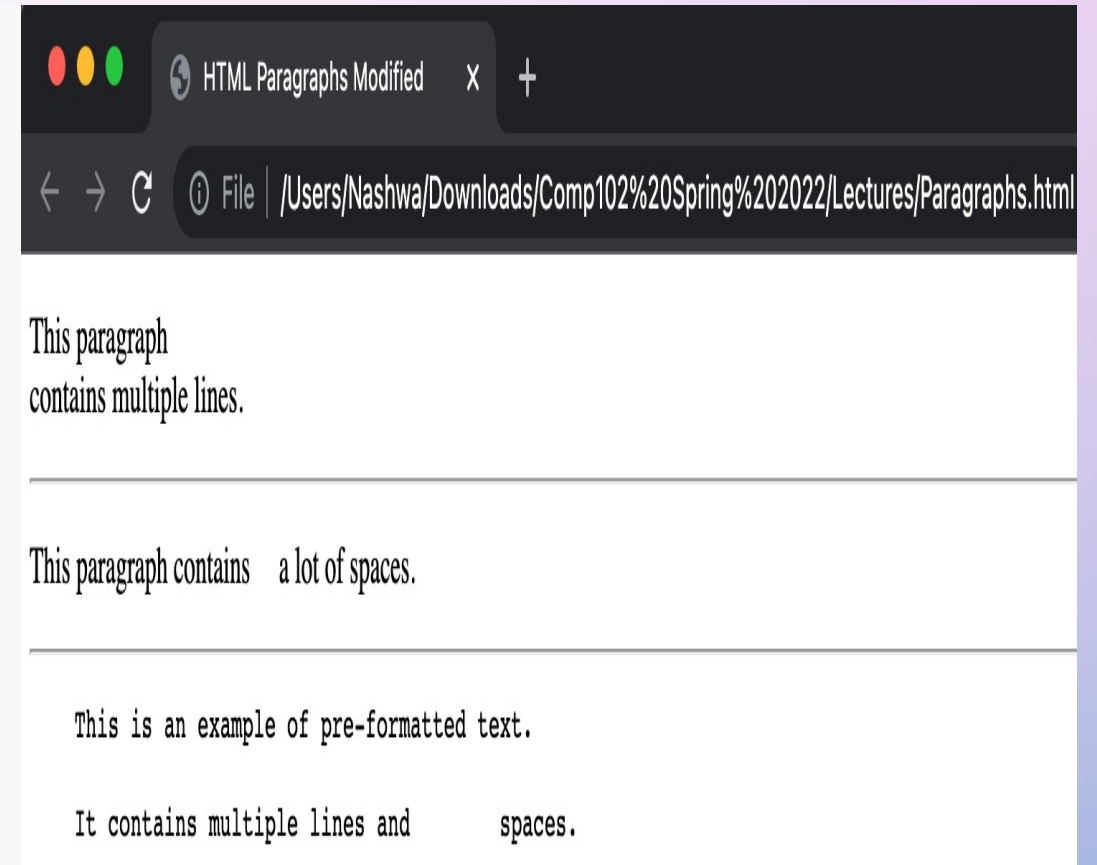
SYNTAX CHARACTERS & SPECIAL CHARACTERS

Character	Description	Entity name	Character	Description	Entity name
<	Less-than symbol	<	‘	Left curly single quote	‘
>	Greater-than symbol	>	’	Right curly single quote	’
"	Quotation mark	"	“	Left curly double quote	“
'	Apostrophe	'	”	Right curly double quote	”
&	Ampersand	&	...	Horizontal ellipsis	…
			©	Copyright	©
			®	Registered trademark	®
			™	Trademark	™
			£	Pound	£
			¥	Yen	¥
			€	Euro	€
			—	En dash	–
			—	Em dash	—

EXAMPLE

```
<body>
  <!--This is a comment -->
  <p>
    This paragraph<br>
    contains multiple lines.
  </p>
  <hr>
  <p>
    This paragraph
    contains&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;a lot of spaces.
  </p>
  <hr>
  <pre>
    This is an example of pre-formatted text.

    It contains multiple lines and      spaces.
  </pre>
</body>
```



ATTRIBUTES

- All HTML elements or tags can have **attributes**.
- Attributes provide additional information about elements and change some of their properties. Some attributes are required, and others are optional.
- Attributes are always specified in **the start tag**.
- Attributes usually come in name/value pairs like: **attributename="value"**.

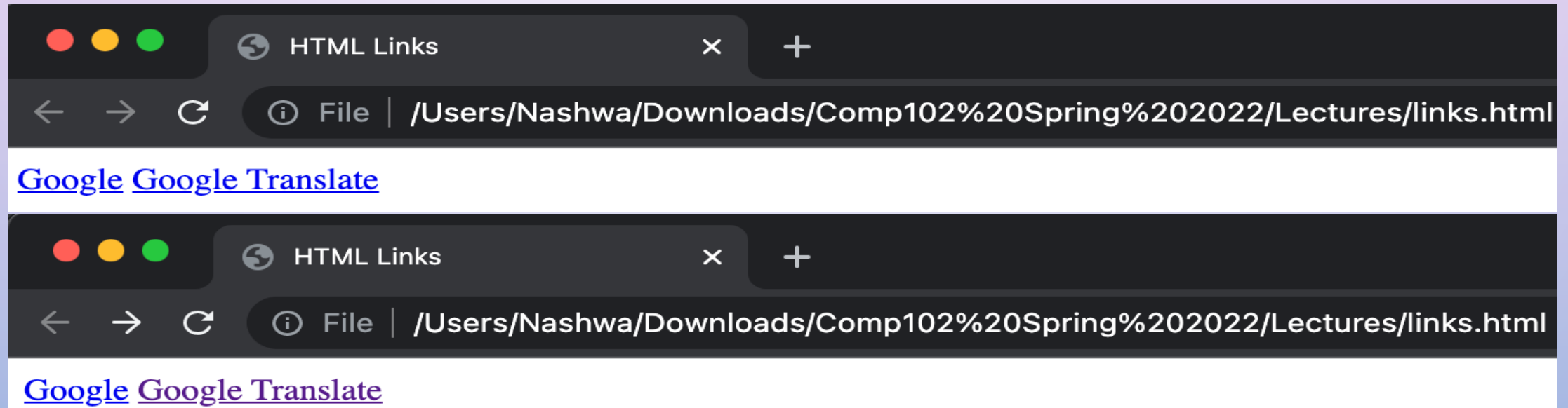
```
<element attributename="value">content</element>
```

HYPERLINKS

- A hyperlink is a connection to another web page.
- When the link is clicked, the corresponding web page is retrieved.
- A link is specified by the anchor tag `<a>` and `</a>`.
- A text that appears within the tag is displayed as a link in the web page.
- The link address is specified through the **attribute "href"**.

HYPERLINKS (EXAMPLE)

```
<body>  
<a href="https://www.google.com">Google</a>  
<a href="https://translate.google.com">Google Translate </a>  
</body>
```



HYPERLINKS (ABSOLUTE VS RELATIVE)

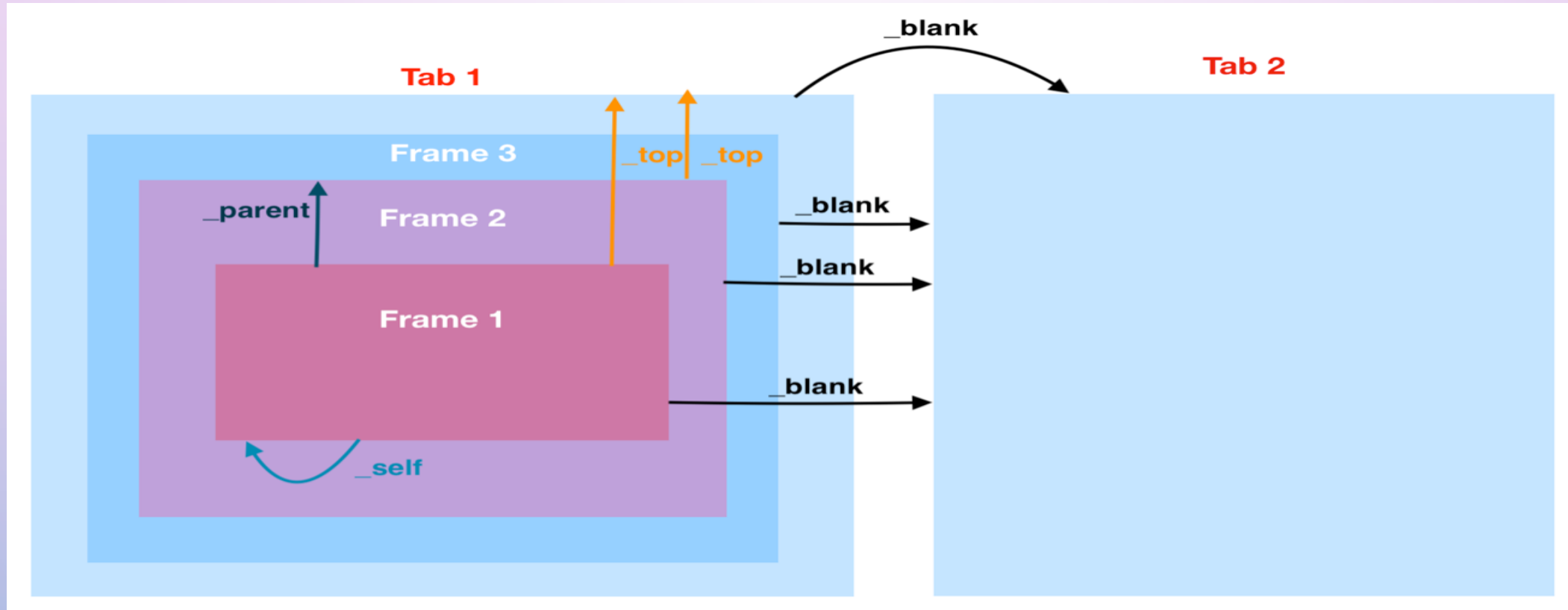
EMAIL LINKS

- If the link specified in the href attribute is a **full web address**, then it is called **absolute**.
- If it is a link to **a local web page** within the same website, then it is called **relative**.
- <https://www.google.com> is an example of an absolute link.
- **“Welcome.Html”** is an example of a relative link.
- To create a link that opens the user's email program, **mailto:** is used in the href attribute.
- For example, `<a href=“mailto:someone@sci.asu.edu.eg”> Contact us</a>`

HYPERLINKS -- TARGET ATTRIBUTE

- The **target** attribute specifies where to open the linked document.
- If the **target** attribute is not specified, then the link will open in the same window or tab, as in the previous example.
- The target attribute has one of the following values:
 - **_self** opens the document in the same tab as it was clicked,
 - **_blank** opens the document in a new tab,
 - **_parent** opens the document in the parent frame,
 - **_top** opens the document in the full body of the window.

TARGET ATTRIBUTE (EXPLAINED)



HYPERLINKS — AT THE SAME WEB PAGE

- Hyperlinks can be used to move to parts (sections) at the same web page.

`<a href="#name">text</a>`

- The link will jump to the section on the page named `name`,
- To create the section, *name*, type in the following HTML code:

`<p id="name">text</p>`

id is an attribute and is not attached to a specific element

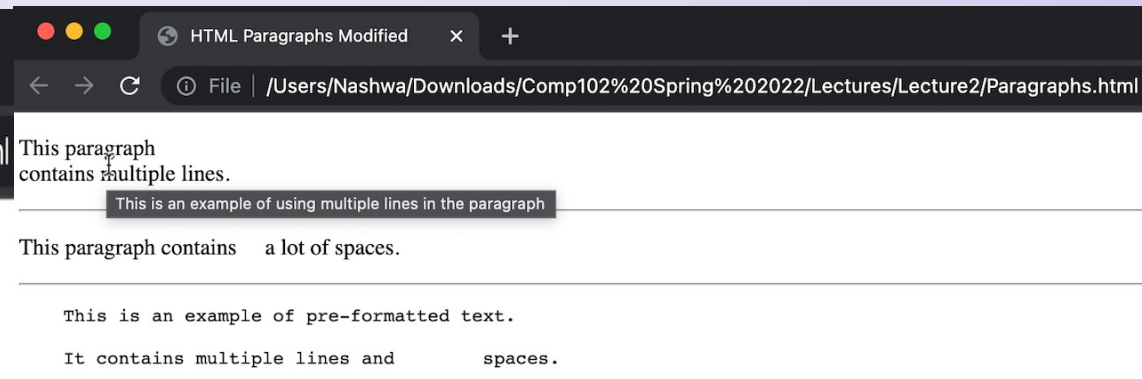
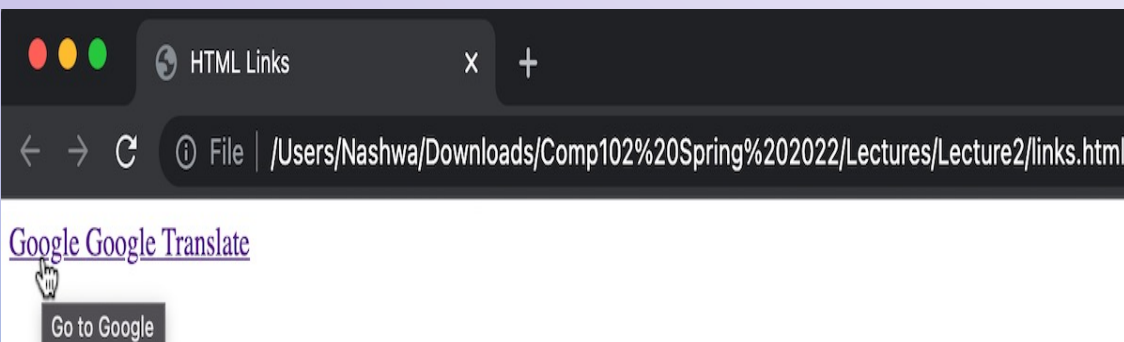
TITLE ATTRIBUTE

- The **title** attribute specifies extra information about an element. The information is most often shown as a tooltip text when hovering over the element.
- This attribute is not used with a specific tag but can be used with any HTML tag.
- For instance, it can be used with the anchor tag **<a>** and the paragraph tag **<p></p>**.

TITLE ATTRIBUTE EXAMPLE

```
<body>
  <a href="https://www.google.com" target="_blank"
  title="Go to Google">Google
</a>
  <a href="https://translate.google.com">Google
  Translate
  </a>
</body>
```

```
<body>  
    <!--This is a comment -->  
    <p title="This is an example of using multiple lines  
in the paragraph">  
        This paragraph<br>  
contains multiple lines.  
    </p>  
    <hr>  
    <p>  
        This paragraph  
contains&nbsp;&nbsp;&nbsp;&nbsp;a lot of  
spaces.  
    </p>
```

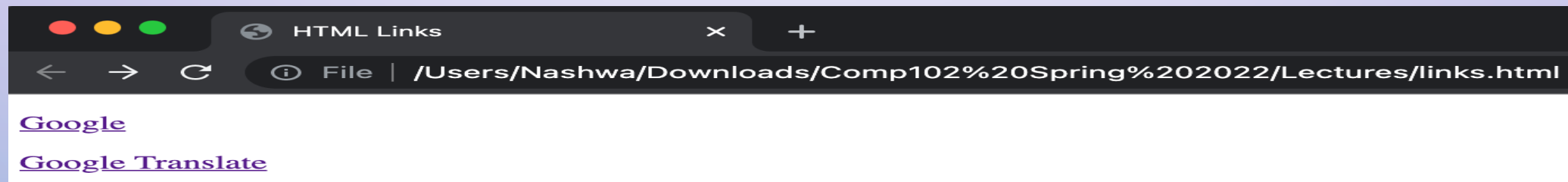


NESTED ELEMENTS

- HTML elements can be nested (i.e., elements can contain elements).
- All HTML documents consist of nested HTML elements.
- `<html><body>` content goes here`</body></html>`.

EXAMPLE

```
<!DOCTYPE html>
<html>
  <head>
    <title>
      HTML Links
    </title>
  </head>
  <body>
    <p><a href="https://www.google.com" target="_blank" title="Go to Google">Google</a>
    </p>
    <p><a href="https://translate.google.com">Google Translate</a>
    </p>
  </body>
</html>
```

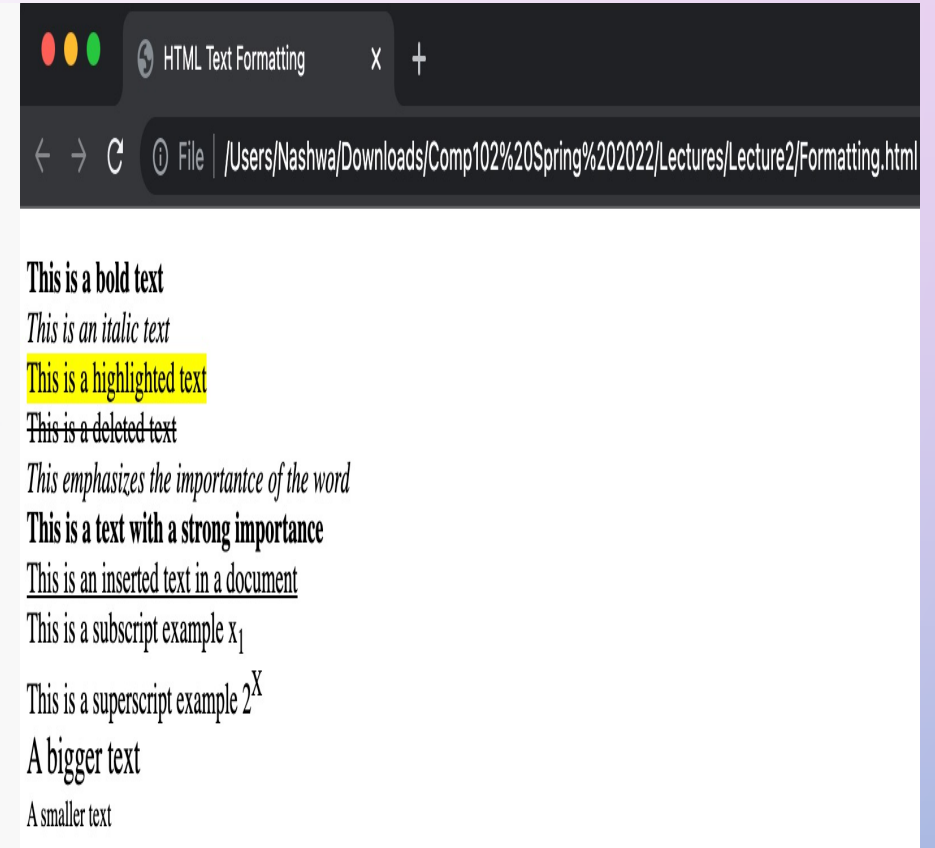


TEXT FORMATTING

<code></code>	defines bold text.
<code><i></code>	defines italic text.
<code></code>	defines text with strong importance.
<code></code>	defines emphasized text.
<code><mark></code>	defines text that should be marked or highlighted.
<code></code>	defines text that has been deleted.
<code><ins></code>	defines a text that has been inserted.
<code><sub></code>	defines subscript text.
<code><sup></code>	defines superscript text.
<code><small></code>	defines smaller text.
<code><big></code>	defines bigger text.

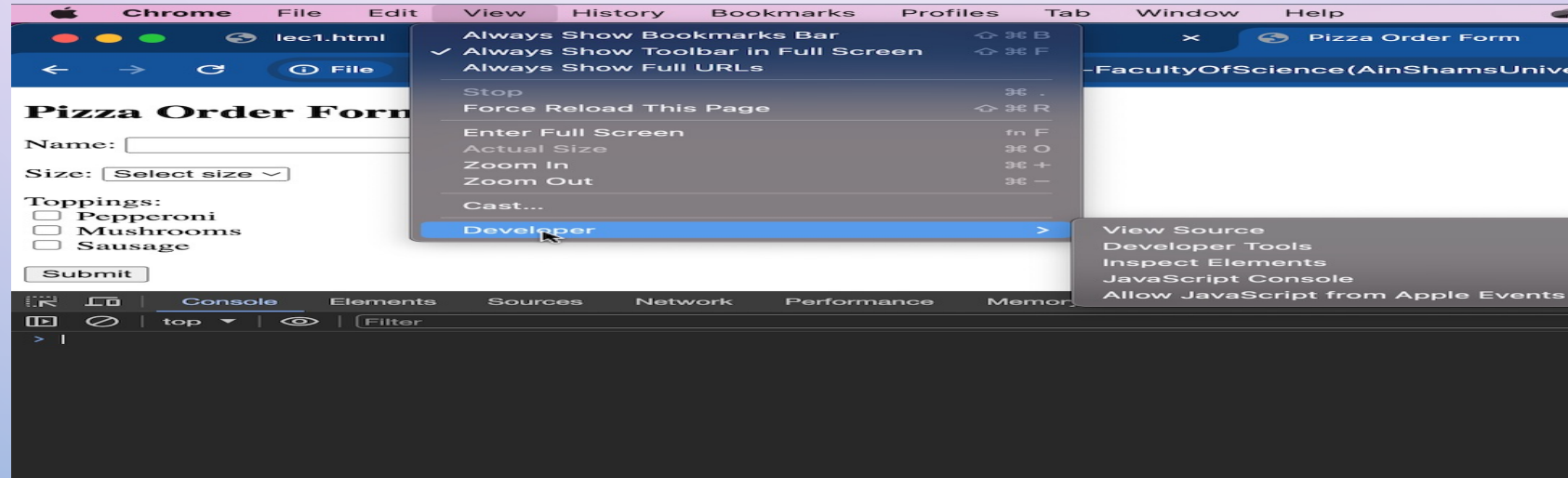
EXAMPLE

```
<body>
  <p>
    <b>This is a bold text</b><br>
    <i>This is an italic text</i><br>
    <mark>This is a highlighted text</mark><br>
    <del>This is a deleted text</del><br>
    <em>This emphasizes the importance of the word </em><br>
    <strong>This is a text with a strong importance</strong><br>
    <ins> This is an inserted text in a document</ins><br>
    This is a subscript example x<sub>1</sub><br>
    This is a superscript example 2<sup>X</sup><br>
    <big>A bigger text</big><br>
    <small>A smaller text</small>
  </p>
</body>
```



VIEW SOURCE & CONSOLE

- To open a source code of any web page, follow the following steps:
 - Right-click on the page and
 - Select "view page source" or "view source".
 - Errors cannot be seen by default in the browser. To see errors and get a lot of other useful information about scripts, “developer tools” have been embedded in browsers.



ORGANIZING PAGE CONTENT

- HTML5 introduced new elements that give semantic meaning to sections of a typical web page or application, including main content (**main**), headers (**header**), footers (**footer**), sections (**section**), articles (**article**), navigation (**nav**), and tangentially related or complementary content (**aside**).
- Use the **main** element to identify the primary content of a page or application.
- The content of a **main** element should be unique to that page.
- In other words, headers, sidebars, and other elements that appear across multiple pages in a site should not be included in the **main** section.

ORGANIZING PAGE CONTENT (CONT.)

- The **header** element is used for introductory material that typically appears at the beginning of a web page or at the top of a section or article.
- There is no specified list of what a **header** must or should contain; anything that makes sense as the introduction to a page or section is acceptable.
- The **footer** element is used to indicate the type of information that typically comes at the end of a page or an article, such as its author, copyright information, related documents, or navigation.
- The **footer** element may apply to the entire document, or it could be associated with a particular section or article.